

CAN Signal Matrix — Display CAN, Pioneer Standard CAN and V2

Status: Working evidence matrix

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Audience: Firmware developer, decoder reviewer and hardware tester

Purpose and evidence levels

This document separates signals by physical bus and vehicle/profile source. It must not be read as proof that similarly named values are interchangeable.

Status	Meaning
Confirmed by MOT	Reproduced during a controlled live capture on the Microlino Pioneer
Plausible	Raw values and behaviour fit the proposed interpretation, but meaning or scale still needs an independent check
Externally supplied as V2	Supplied by a separate reverse-engineering source under the V2 label, but the exact vehicle, battery and BMS generation are not documented
Externally indicated	Present in supplied notes/screenshots, but byte order, scale, signedness or exact semantics remain incomplete
Not observed	Targeted ID did not appear during the stationary Pioneer capture
Observed on large-battery vehicle	Seen by MOT on a newer Microlino with the 200-240 km battery; interpretation still requires controlled state changes
Confirmed for large-battery decoder	Original PID/Flespi material agrees with physically plausible values in a 16,675-frame capture from the newer large-battery vehicle

All MOT captures used two ESP32-C6 TWAI controllers at 500 kbit/s in listen-only mode. CAN1 was connected to Standard CAN and CAN2 to Display CAN.

Cross-bus signal summary

Signal	Display CAN	Pioneer Standard CAN (V1 vehicle)	Externally supplied / newer large-battery vehicle	Current assessment
Display SOC	0x602 data[0] / 2	No equivalent confirmed	External source: 0x18D data[7]; large-battery live frame: 0x05 with meaning unknown	Display CAN confirmed by existing decoder. The external V2 assignment cannot be generalized to the tested large-battery vehicle.
Standard/BMS SOC candidate	Not used	0x18D data[7]	External source: same byte; large-battery semantics unknown	Raw Pioneer field confirmed, semantics unresolved. Keep every source and vehicle generation separate.
Speed	0x602 data[1] / 2	Not found	No supplied mapping	Display-CAN decoder field.
Odometer	0x602 data[4..7] with current packed scaling	Not found	No supplied mapping	Display-CAN decoder field.
HV pack voltage	Not currently decoded	0x18D data[3..4], unsigned little-endian mV	External source: same mapping; large-battery frame decodes to impossible 792 mV	Confirmed only on Pioneer. The supplied mapping is unusable on the tested large-battery vehicle.
Battery current	Not currently decoded	0x18D data[1..2], signed little-endian; raw x 0.3 A	External material lists 0x2BA; old layout yields impossible 1,306.2 A on the large-battery frame	Scale and sign confirmed only on Pioneer. 0x2BA is a discovery target for the newer vehicle.
DC charging power	Dashboard power byte is unscaled and bidirectional	Derived as pack voltage × battery current	Not derivable until voltage and current mappings are verified	Pioneer result is confirmed; the current V2 decoder must not publish the same calculation as valid for the large-battery vehicle.
Plugged	Existing candidate 0x604 data[4] & 0x10	0x18D data[6]: stable 0x10 unplugged and 0x20 plugged	External source calls byte 6 charge/status; large-battery live value 0x03	Pioneer Standard-CAN state confirmed. The stable plugged value or bit for the newer vehicle is unknown.
Active charging	Current 0x603 power-threshold inference is not reliable	Positive signed 0x18D data[1..2]; idle offset about -3/-4	Unknown on the large-battery vehicle	Confirmed only on Pioneer across start, stable charge, AC removal and natural completion.
Cell values A/B	Not decoded	0x4AD data[0..1] and [2..3], unsigned little-endian mV	External source: same mapping; 0x4AD absent on tested large-battery vehicle	Plausible on Pioneer. External vehicle assignment is unresolved and the mapping is unusable on the tested vehicle.
Cell min/max/delta	Not decoded	Derived from the two 0x4AD values	External source: same derivation; unavailable without 0x4AD on large-battery vehicle	Plausible only for the observed Pioneer payload.
SOH	Not decoded	Not found	0x1B1 data[2..3] / 100 candidate	Externally indicated only.

Signal	Display CAN	Pioneer Standard CAN (V1 vehicle)	Externally supplied / newer large-battery vehicle	Current assessment
Pedal/power request	0x603 data[4] is the dashboard power display	0x37F data[2..3] candidate	Same candidate	0x37F was present but constant in stationary Pioneer testing; driving evidence is required.

Pioneer Standard-CAN findings

CAN ID 0x18D

Bytes	Type and conversion	Proposed meaning	Pioneer evidence	Relation to externally supplied V2 data
data[0]	uint8	rolling counter or sequence	Covered nearly the full byte range during stable charging; not a direct measurement	Unknown
data[1..2]	int16, little-endian	battery current; amperes = raw x 0.3	Scale and sign confirmed during charging, traction and regeneration	Likely related, but the V2 source does not expose an unambiguous raw-ID association
data[3..4]	uint16, little-endian mV	HV pack voltage	Confirmed by MOT	Same mapping was supplied, but it fails on the tested large-battery vehicle
data[5]	uint8	unknown	Stable 0x00 in recorded phases	Unknown
data[6]	uint8 state	0x10 unplugged; 0x20 plugged	Confirmed by repeated powered and AC-deenergized plug tests; transient 0x04 also occurred	V2 source calls it a charge/status byte
data[7]	uint8	Standard/BMS SOC candidate	Raw value confirmed; stayed 90 while Pioneer display showed 97–100%	Same byte was supplied as Display SOC; large-battery live value 0x05 remains uninterpreted

Charging evidence for 0x18D data[1..2]

Vehicle/charger state	Signed raw	Pack voltage	External measurement	Interpretation
Awake, unplugged	about -4	about 53.0 V	0 W	Idle offset
Plugged, EVSE AC-side off	about -4	about 53.0 V	0 W	Plugged but not charging
Stable charge	112–118	about 53.3 V	1,836 W gross AC input	$115 \times 0.3 \text{ A} \times 53.3 \text{ V} \approx 1,839 \text{ W}$
Taper charge at 98–99% display SOC	43–47	about 53.2 V	762 W gross AC input	$45 \times 0.3 \text{ A} \times 53.2 \text{ V} \approx 718 \text{ W}$ before conversion losses
External AC removed, cable retained	about -4	falling from charge voltage	0 W	Charging stopped, still plugged
Natural completion at 100% display SOC	fell from +45 to -3	about 53.24 V at end	Fell from 762 W to 0 W	Stable post-charge idle offset

Lower-SOC measurements at several charge settings confirmed 0.3 A per raw unit. At about 50 V, the derived DC values tracked independent 1,809 W, 2,088 W and approximately 2,340 W references within expected conversion and vehicle-load differences. Three controlled road captures then confirmed negative current during traction and positive current during regeneration. Observed results included about 4.8 kW pedal-release regeneration, 7.8 kW with light braking and a later normal-road peak of about 20.9 kW discharge.

CAN ID 0x4AD

Bytes	Type and conversion	Proposed meaning	Status
data[0..1]	uint16, little-endian mV	cell value A	Plausible on Pioneer; externally supplied under an unresolved V2 label
data[2..3]	uint16, little-endian mV	cell value B	Plausible on Pioneer; externally supplied under an unresolved V2 label
derived	$\min(A, B)$	minimum cell value	Plausible
derived	$\max(A, B)$	maximum cell value	Plausible
derived	$\max(A, B) - \min(A, B)$	cell delta	Plausible

Stable examples included 4,081/4,078 mV before charging and approximately 4,105/4,100 mV during charge. Larger transient divergence occurred around charge start, so the exact semantics must remain provisional until repeated captures exclude multiplexing or transition-specific payload use.

Newer Microlino with large battery

A live Standard-CAN capture from a newer Microlino with the approximately 200-240 km battery received hundreds of thousands of CAN1 frames without bus errors. 0x18D was present continuously, while 0x4AD was absent. One stable 0x18D payload was:

01 02 11 18 03 00 03 05

Applying the Pioneer layout would produce 1,306.2 A, 792 mV pack voltage, status 0x03 and an SOC candidate of 5%. The current and voltage results are physically impossible, proving that the Pioneer byte layout cannot be used for this vehicle. The frame must remain raw discovery data until controlled unplugged, plugged, charging, traction and regeneration captures identify the actual signals.

The earlier source may have been recorded from a Pioneer/V1 vehicle, or from a newer Microlino whose medium battery and BMS retain the Pioneer layout. This is a working hypothesis only: the source does not document the exact vehicle, battery or BMS generation. The name V2 must therefore not be treated as a single CAN protocol generation.

Confirmed large-battery BMS layout

The recovered original PID workbook and Flespi configuration identify a separate big-endian layout. A 16,675-frame capture from the large-battery vehicle contains all three IDs with matching physically plausible values:

CAN ID	Direct bus decoding	Captured evidence	Decoder status
0x1B0	bytes 2-3 pack mV, 4-5 minimum cell mV, 6-7 maximum cell mV; big-endian	47,317-47,326 mV pack and 3,635-3,643 mV cells	Implemented
0x1B1	bytes 0-1 SOC x100, bytes 2-3 SOH x100; big-endian	32.15% SOC and 97.01% SOH	Implemented as BMS-specific values
0x2BA	bytes 0-1 signed big-endian current x10	stationary -1.9 to -2.1 A	Implemented; sign still needs drive/charge confirmation

The direct CAN order must be used by MOT. The Flespi manual .can .hex . * conversion reverses groups of 16-bit words internally and must not be copied as bus byte order. Pack power is derived only when fresh 0x1B0 voltage and plausible 0x2BA current are available. Plugged and active charging remain unknown. AWS continues to publish Display-CAN 0x602 as display/soc; BMS SOC is kept separate and never overwrites the confirmed display value.

Other targeted Pioneer IDs

CAN ID	Pioneer result	External V2 information
0x101	Not observed	Byte 4 candidate for positive power request; unit/scale incomplete
0x107	Not observed	Separate ESPHome source watched bytes 0–3 without a decoding rule
0x1B0	Not observed	Confirmed big-endian pack/min/max layout on the large-battery vehicle
0x1B1	Not observed	Confirmed big-endian SOC/SOH x100 layout on the large-battery vehicle
0x2BA	Not observed	Signed big-endian current /10 implemented; dynamic sign validation pending
0x37F	Observed and constant while stationary	Bytes 2–3 pedal/power-request candidate

Display-CAN signal map

CAN ID 0x602

Bytes	Current decoding	Status
data[0]	SOC = byte / 2	Implemented Display-CAN signal
data[1]	speed = byte / 2 km/h	Implemented Display-CAN signal
data[4..7]	packed odometer / 1024 km	Implemented Display-CAN signal
derived	estimated range from SOC	Firmware-derived estimate, not a direct vehicle signal

CAN ID 0x603

The external source uses one-based byte and bit numbering. Positions below are converted to zero-based firmware indices, assuming its bit 1 is the least- significant bit.

Firmware position	Mask	Meaning	Status
data[1]	0x01	seat-belt sensor/status	Externally indicated
data[2]	0x08	handbrake	Externally indicated

Firmware position	Mask	Meaning	Status
data[2]	0x02	parking/position light	Externally indicated
data[2]	0x04	low beam	Externally indicated
data[3]	0x80	D/R interlock warning when brake was not pressed first	Externally indicated
data[3]	0x20, 0x40	indicator-state bits	Externally indicated; left/right mapping missing
data[4]	full byte	dashboard power display	Implemented; sign/scale and charging interpretation remain provisional
data[5]	0x01	sport mode	Externally indicated
data[6]	0x02	brake pressed	Externally indicated

data[4] is a bidirectional dashboard display value. It must not by itself be used as proof of active charging. The current firmware threshold-derived charging state is compatibility behaviour pending replacement by the confirmed Standard- CAN current/plug signals or another independently verified source.

CAN ID 0x604

Firmware position	Mask	Meaning	Status
data[4]	0x10	plugged candidate	Implemented, but not yet directly compared with the controlled plug sequence

Decoder policy

- Preserve Display-CAN and Standard-CAN values as source-specific telemetry.
- Do not overwrite Display-CAN SOC with 0x18D data[7] on Pioneer.
- Permit every registered decoder profile on every physical CAN controller; board defaults are configuration policy, not decoder-engine restrictions.
- Keep CAN operation listen-only during discovery and pilot operation.
- Promote plausible or externally indicated fields to confirmed only after a repeatable state change or independent physical reference.

Remaining validation

- Stationary/drive comparison of 0x37F to test the pedal hypothesis.
- Direct Display-CAN 0x604 plug-state comparison.
- Compare 0x2BA during unplugged, plugged-idle, charging, traction and regeneration states and identify the large-battery plug/charge signal.
- Identify the exact vehicle, battery and BMS generation behind the externally supplied 0x18D/0x4AD material.
- Controlled cross-channel injection and extended dual-CAN soak.

Related documents

- CAN and decoder pipeline
- Firmware architecture
- C6-001 dual-CAN sprint